

STRATEGIC ANALYSIS FOR THE COUNTRY OPERATIONS BUSINESS PLAN 2016–2018

I. DEVELOPMENT TRENDS AND ISSUES

A. Country Background

1. Palau is a small island nation located in the western Pacific Ocean. Palau achieved political independence in 1994 under the Compact of Free Association with the United States (US).¹ The country's population of around 21,000 is spread across 250 islands, most populous island is Koror. Since the early 2000s, tourism related activities have rapidly grown reflecting active foreign investment. The country is now one of the major tourist destinations in the Pacific. The main productive sectors of the economy are tourism related services (hotels, restaurants and transport), subsistence agriculture, and fishing. Government is a major employer. The Compact of Free Association brings with it substantial financial assistance from the United States, which supports government expenditure.

2. **Economic performance.** A surge in tourism and increased construction drove gross domestic product (GDP) growth by 6.9% in FY2014, reversing the economic contraction of the previous year. Following a 9.5% decline in FY2013, visitors arrivals rebounded by 13.2%, mainly from the People's Republic of China, to reach a record high of nearly 123,000. New air links with Hong Kong, China and expanded accommodation capacity boosted the recovery in tourism arrivals. Palau recorded a small fiscal surplus of 1.2% of GDP in FY2014, making it the fourth consecutive year Palau has recorded a positive fiscal balance. Reductions in grant receipts were more than offset by higher tax collections.

3. The Palau economy is expected to retain its strength backed by the strong tourism growth, hotel construction, higher public infrastructure spending, and improvements in the external economic environment. Tourism stands to benefit from lower air transport costs as visitors arrivals continue to climb at double-digit rates. However, serious concerns have been growing about the country's capacity to accommodate the increasing tourist numbers. Major problems include the lack of hotel rooms, insufficient infrastructure, and environmental degradation. With the limited capacity, the country has been struggling to find ways to move to high end tourism.

4. **Development challenges.** Palau's main development challenges include (i) creating an environment conducive to high end tourism; (ii) making growth more inclusive; and (iii) consolidating fiscal balances. Despite having a vision focused on high end tourism, Palau has not yet attained the goal. Average tourist spending remains around \$500 per day, far below the over \$1,000 a day in the Maldives – a comparable small-island economy that is also highly dependent on tourism. A recent sharp increase in tourist arrivals has highlighted Palau's limited capacity in basic infrastructure, tourism-related services, and environmental protection. Palau needs a strategic approach to upgrading its tourism sector in keeping with its capacity. A growing tourism industry has not translated to a corresponding increase in job opportunities for the local workforce. The majority of jobs in hotels and related services are filled by foreign workers while locals continue to migrate, mainly to Guam, Hawaii and mainland US, in search of better job opportunities and higher wages. A large portion of Palau's national budget is still covered by US assistance through the Compact. Given the planned decrease in Compact support, long-term fiscal sustainability is a key challenge. Fiscal consolidation requires actions both in revenue (a comprehensive tax reform) and expenditure (wage bill control). However, progress has been slow, as reflected in the pending tax reform bill at the congress.

¹ The Compact of Free Association defines the relationship that Palau has entered into with the United States, and includes the provision of economic assistance, including grants.

5. **Millennium Development Goals.** Palau is on track to meet the Millennium Development Goals (MDG) relating to achieving universal primary education (goal 2), promoting gender equality and empowering women (goal 3), reducing child mortality (goal 4), improving maternal health (goal 5), combating HIV/AIDS, malaria and other diseases (goal 6) and ensuring environmental sustainability (goal 7). Mixed progress is being made in eradicating extreme poverty and hunger (goal 1)..

6. **Poverty.** Poverty estimates are dated based on the Household Income and Expenditure Survey in 2005/2006. This shows that 18.4% of households and 24.9% of population were below the national poverty line (basic needs).² While rural areas had a slightly higher poverty incidence (20.8% of household and 28.9% of population), there was not big difference with urban areas (19.2% and 26.2% respectively). The central cause of poverty remains the lack of job opportunities. The local work force has limited access to the jobs that are created by tourism- related businesses.

7. **Debt Sustainability.** Palau is a Group B country and has access to both the Asian Development Fund and ordinary capital resources. The first debt sustainability assessment (DSA) for Palau was conducted as a part of IMF Article IV consultation in 2014. The DSA indicates that the country's future debt path would remain sustainable. However, it also shows that the debt risk could increase without fiscal reforms, calling for continued fiscal consolidation and prudent debt management.

B. Highlights of Previous Asian Development Bank Country Strategy

8. ADB has provided technical assistance to Palau to strengthen national development planning and economic and fiscal management. ADB helped develop a medium-term budget framework that has enabled the government to plan more effectively, to improve tools for fiscal management and deficit reduction, and to facilitate the implementation of development plans. ADB has built upon these initiatives by assisting the government in preparing the Medium-Term Development Strategy.

9. To improve access to business credit, ADB has provided technical assistance for new secured transaction legislation. The government has launched a secured transaction registry, making it possible for banks to provide secured loans so reducing their risk. ADB has also supported the government to improve governance in state-owned enterprises (SOEs), develop an SOE policy that will eventually include relevant legislative measures, and facilitate business growth by improving business registration systems.

10. A country environment assessment has been prepared, and ADB has provided support for an action plan on climate change, implemented through a pilot project in Melekeok. Through the various regional programs it supports, ADB will investigate opportunities to further support climate resilience in Palau.

11. ADB support for sustainable health care financing – originating in 2008 to establish a medical savings plan and a health insurance scheme – was extended in 2013, with ADB providing assistance for the cost accounting of all health services in Palau. Also in 2013, ADB released the second tranche of a program loan for a series of technical and infrastructure reform activities to support Palau's economic growth, public health, and environmental protection, and to improve basic water and sewerage services. ADB approved its first investment project for

² ADB is supporting preparation of the 2015/2016 Household Income and Expenditure Survey.

Palau, a loan to upgrade the sewerage network in Koror and Airai, which will directly benefit about 70% of the Palauan population, while helping to cater to a growing tourism market.

12. In 2014, ADB provided support for Palau to conduct the Households Income and Expenditure Survey, and to prepare the joint North Pacific Regional Connectivity Project with the Federated States of Micronesia, which will provide broadband internet access via deep-sea fiber optic cabling.

II. THE COUNTRY STRATEGIC PRIORITIES

A. Developing the Asian Development Bank's Country Strategic Priorities

13. The current administration when it took office in January 2013 adopted the Management Action Plan (MAP), based largely on the Medium-Term Development Strategy. The overarching goal of the MAP is improving the quality of life of the people of Palau, while underscoring commitment to environmentally sustainable and social inclusive development. To this end, priorities are (i) the diversification of the economy and investments in human resources and infrastructure; (ii) private sector and public-private service provision, investment, and economic development; (iii) youth and their opportunities for education, employment, and participation in the development process; (iv) universal access to effective health care; and (v) the integration of traditional values and environmental considerations in decision making and development planning.

14. The country operations business plan, 2016 – 2018 for Palau aligns with ADB's Interim Pacific Approach 2015, and with the Midterm Review of Strategy 2020.³ The COBP aligns with the government's Medium Term Development Strategy as well as the MAP. Through the COBP, ADB will continue to support government's priorities on (i) improving public sector effectiveness, (ii) facilitating private sector development, (iii) delivering safe water and sanitation services, and (iv) improving connectivity.

15. The United States is the main development partner of Palau. ADB, Australia, the European Union (EU), Japan, Taipei,China, the United Nations Development Programme and the World Bank also provide assistance to Palau. US federal grant programs provides support across all sectors with the major focus on education, health and infrastructure. Australia's focus is on capacity development for the education, health and public sectors, while the European Union focuses on energy and environment sector development especially on renewable energy. Japan supports energy, environment, education and health as well as waste management and water. Taipei,China focuses its development assistance on infrastructure, agriculture and environment; likewise, UNDP also focuses on the environment. The World Bank has no projects in Palau thus far; however, it is providing technical assistance towards regulatory reforms of the ICT sector. The International Monetary Fund through the Pacific Financial Technical Assistance Centre is providing support for tax reform in reviewing the present tax strategic and operational directions.

16. In November 2013, the government, through its Economic Advisory Group, comprising members of the public and private sector, hosted the first annual economic summit for Palau. The objective of the summits has been on the implementation of the Medium Term Development Strategy, with a particular focus on tourism and other priorities of the government. The summit is well attended by key stakeholders in both the public and private sector as well as development partners and ADB has provided support through technical resource persons for

³ The Interim Pacific Approach, 2015 extends the validity of The Pacific Approach 2010-2014, which serves as the country partnership strategy for Palau.

the summit. The next summit will be held in November 2015.

B. Implementation Issues

17. Since joining ADB in 2003, Palau has received two loans and five technical assistance projects. Additionally Palau has benefitted from regional TA projects supporting public sector management, education, economic management and private sector development. The first investment project for Palau was approved in late 2013 so implementation is underway. An emerging challenge is the capacity of the government to implement large projects, both self-funded and development partner-supported. Therefore, no lending pipeline is proposed for 2016–2018 as the government will focus on implementing the Koror-Airai Sanitation Project and the proposed North Pacific Regional Connectivity Project (to be approved in 2015) over the next few years.

18. ADB has strengthened its in-country presence with the appointment of a local coordinator in 2012. The main role of the local coordinator is to facilitate the effective partnership between Palau and ADB, as well as between ADB and its development partners. The role is currently funded under TA, and securing long-term funding for this position is important to the ongoing sustainability of ADB's operations in Palau.

Data Tables

Table 1: Millennium Development Goals

Millennium Development Goal	General Assessment in 2008 ^a	Assessment in 2013 ^b
Goal 1: Eradicate Extreme Poverty and Hunger	Mixed	Mixed
Goal 2: Achieve Universal Primary Education	On Track	On Track
Goal 3: Promote Gender Equality and Empower Women	Mixed	On Track
Goal 4: Reduce Child Mortality	On Track	On Track
Goal 5: Improve Maternal Health	On Track	On Track
Goal 6: Combat HIV/AIDS, Malaria and Other Diseases	Mixed	On Track
Goal 7: Ensure Environmental Sustainability	On Track	On Track
Goal 8: Develop a Global Partnership for Development	Mixed	Not assessed

^a United Nations Development Program. 2009. *Millennium Development Goals. Initial Status Report 2008*. Koror. Pacific Islands Forum Secretariat. 2013. *2013 Pacific Regional MDGs Tracking Report*. Suva.

Table 2: Human Development Indicators

	HDI value	HDI rank	Life expectancy at birth (years)	Mean years of schooling (years)	Expected years of schooling (years)	GNI per capita (constant 2011 PPP \$)
2013	0.775	60	72.4	12.2	13.7	12,823

GNI = gross national income, HDI = human development index, PPP = purchasing power parity.

Source: United Nations Development Program. Human Development Statistical Tables.

Table 3: Poverty Incidence 2005/2006, percent, (Basic Needs)^a

	Households	Population
Palau (average)	18.4	24.9
Urban	19.2	26.2
Rural	20.8	28.9

^a Basic Needs Poverty Line , which includes an allowance for essential non-food expenditure, was estimated as a national average expenditure per household per week.

Source: The Government of Palau and UNDP, 2008. *Analysis of Household Income and Expenditure Survey 2005/2006*. Suva.

Table 4: 2014 Environment Country Performance Assessment Indicators

	Score ^a
PART A: Institutional Context	
Access to information	3.00
Public participation	5.00
Quality and effectiveness of EA system	4.00
Cross-sectoral coordination	4.00
Accountability	4.00
Subtotal	4.00
PART B: Environmental Themes	
Air Pollution Management	3.00
Water pollution management	4.00
Solid and hazardous waste management	4.00
Freshwater resources	4.00
Marine and coastal resources management	5.00
Ecosystem and biodiversity management	5.00
Commercial renewable resources management	4.00
Commercial non-renewable resources management	3.00
Climate change	3.00
Subtotal	3.89
Grand Total	4.00

^a Highest score for each category is 6.0.

Source: Asian Development Bank.

Table 5: 2014 Country Performance Assessment Ratings^a

CRITERIA	PAL	Pacific	Group A ^b	Group B ^c
A. Economic Management	3.3	3.3	4.1	4.4
1. Macroeconomic Management	3.5	3.4	4.3	4.4
2. Fiscal Policy	2.5	3.1	4.0	4.3
3. Debt Policy	4.0	3.4	4.1	4.5
B. Structural Policies	3.3	3.3	3.8	4.1
4. Trade	4.0	4.0	4.3	4.1
5. Financial Sector	3.5	3.0	3.4	4.0
6. Business Regulatory Environment	2.5	2.8	3.6	4.2
C. Policies for Social Inclusion/Equity	3.6	3.2	4.1	4.2
7. Gender Equity	3.0	3.1	4.0	4.1
8. Equity for Public Resource Use	3.0	3.3	4.3	4.5
9. Building Human Resources	4.5	3.2	4.3	4.4
10. Social Protection and Labor	3.5	3.1	3.9	4.1
11. Policies and Institutions for Environmental Sustainability	4.0	3.0	3.9	3.8
D. Public Sector Management and Institutions	3.4	3.2	3.8	4.0
12. Property Rights and Rules-based Governance	4.0	3.3	3.4	3.9
13. Quality of Budgetary and Financial Management	3.5	3.3	3.9	4.4
14. Efficiency of Revenue Mobilization	3.5	3.4	4.1	4.3
15. Quality of Public Administration	3.0	3.0	4.0	3.9
16. Transparency, Accountability and Corruption in the Public Sector	3.0	3.1	3.5	3.5
E. Portfolio Performance	4.0	3.5	3.3	3.3
17. Portfolio Performance	4.0	3.5	3.3	3.3

PAL = Republic of Palau.

^a Highest score for each category is 6.0.

^b Group A countries are: Afghanistan, Bhutan, Cambodia, Kyrgyz Republic, Lao People's Democratic Republic, Maldives, Nepal, and Tajikistan.

^c Group B countries are: Armenia, Bangladesh, Georgia, Mongolia, Pakistan, Sri Lanka, Uzbekistan, and Viet Nam.

Source: Asian Development Bank.

Table 6: Palau Country Economic Indicators

Item	Fiscal Year				
	2011	2012	2013	2014 ^e	2015 ^p
A. Income and Growth					
1. GDP per capita (\$, current)	9,766.1	10,443.5	10,957.3	11,748.3	12,117.7
2. GDP growth (% , constant prices)	7.7	4.0	(1.7)	8.0	10.0
a. Agriculture	9.6	1.6	(8.0)	(1.8)	1.8
b. Industry	5.3	(5.0)	(15.8)	7.4	33.2
c. Services	7.0	4.6	(0.3)	7.0	6.0
B. Saving and Investment (% of GDP, current and market prices)					
1. Gross domestic investment	25.6	24.6	20.7	18.3	26.2
2. Gross domestic saving	...	...	...	...	...
C. Money and Inflation (annual % change)					
1. Consumer price index	2.6	5.4	2.8	4.0	4.0
2. Total liquidity (M2)	...	...	...	...	...
D. Government Finance (% of GDP)					
1. Revenue and grants	44.1	44.7	41.0	41.2	40.6
2. Expenditure and onlending	42.9	43.7	39.7	40.0	37.4
3. Overall fiscal surplus (deficit)	1.2	1.0	1.2	1.2	3.1
E. Balance of Payments					
1. Merchandise trade balance (% of GDP)	(56.0)	(54.9)	(55.0)	(51.6)	(51.3)
2. Current account balance (% of GDP)	(13.1)	(10.1)	(8.8)	(5.1)	(6.4)
3. Merchandise export (\$) growth (annual % change)	5.9	18.8	(8.1)	33.8	(30.9)
4. Merchandise import (\$) growth (annual % change)	19.5	6.8	4.2	4.6	(1.2)
F. External Payments Indicators					
1. Gross official reserves (including gold, \$ million in weeks of current year's imports of goods)	...	...	...	...	...
2. External debt service (% of exports of goods and services)	5.7	5.1	4.9	4.4	5.0
3. External debt (% of GDP)	31.1	31.9	28.4	27.1	24.2
G. Memorandum Items					
1. GDP (\$ million, current prices)	201.6	216.7	229.0	247.1	256.5
2. Exchange rate (\$/\$, average)	1.0	1.0	1.0	1.0	1.0
3. Population (million)	20.6	20.8	20.9	21.0	21.2

... = data not available, GDP = gross domestic product, M2 = a classification of money supply.

^a Estimated.

^b Projected.

Sources: ADB. Asian Development Outlook database; International Monetary Fund Article IV Consultation Staff Reports (various years); Republic of Palau FY2014 Economic Review; and Palau Office of Planning and Statistics.

SECTOR ASSESSMENTS

A. Water and Other Urban Infrastructure Services

1. Sector Performance, Problems, and Opportunities

19. **Water resources.** Most water supply systems in Palau rely on surface water sources, including the Koror–Airai water supply system, which relies on the Ngerikil and Ngerimel rivers. Babeldaob Island has 15 small water systems. One has a groundwater supply source while all the others use stream flow. Storage volumes in tanks and at the low head dams for stream intakes are small and not sufficient to substantially supplement stream shortfalls during dry periods. As a result, the stream sources must provide the system’s full consumptive use on a daily basis.

20. Experience since 1990 indicates that the reliable yield of the current water sources for Ngerikil and Ngerimel rivers is substantially less than the current demand of 3.8 million gallons/day⁴ in Koror and Airai. Stream flow analysis indicates the reliable yield of the existing Koror–Airai water supply system is approximately 1.0 million gallons/day during a 25-year drought (a drought that is expected to occur once in 25 years). Disruptions to the water supply, affecting approximately 14,000 people, are experienced during severe droughts, as in 1983, 1997, and 2003. The government is concerned that the frequency of severe droughts may increase due to climate change, and has identified the need to improve water supply security for the Koror–Airai area.

21. **Anticipated water supply shortfall.** Demand for water from the Koror–Airai water supply system is rapidly approaching the water production capacity which is forecast to increase demand to 4.4 million gallons/day by 2020. The existing production capacity of the Koror–Airai water supply system is 4 million gallons/day. Demand for treated water will exceed supply by 2015 unless measures are implemented to reduce system losses and the high per capita demand.⁵

22. **High water use.** The Koror–Airai water supply system produces an average of 3.8 million gallons/day of treated water. This equates to a daily per capita production rate of approximately 230–240 gallons.⁶ This rate is higher than in any other Pacific country (typically 120–150 gallons/day) and higher than Asian countries by factors of 5–10. Even with the high system losses, per capita demand for water is very high and is attributed to limited consumer appreciation of the value of water and inadequate tariffs.

23. **System fragmentation and inefficiency.** The national water supply systems, including water treatment plants, have been substantially improved over the past 15 years and the water supply networks expanded, particularly in Airai state. However, operations are still inefficient: losses are high (estimated at about 43%). Unmetered connections remain and a flat fee tariff is applied for water and sewer until 100% metering is in place. A program is in place to achieve this.

24. Until recent, the Koror–Airai and water supply systems and the Koror and Melekeok sewage systems were operated and maintained by the Bureau of Public Works (BPW). In July 2013, Palau Public Utilities Corporation Water and Wastewater Operations became responsible

⁴ 1 US gallon is 3.785 liters.

⁵ The forecast demand for 2020 is based on a medium-growth scenario for population and tourist arrivals.

⁶ Due to system losses, the production of 908 liters/ day equates to consumption of 632 liters/day.

for the delivery of these services. The water system services about 98.9% of the urban population and 86.2% of the rural population—or 96% of the total population.

25. **Sanitation Services.** Koror has a centralized sewerage system which services more than 80% of the Koror population. The network comprises 40 kilometers (km) of gravity mains, 13 km of force (pumped) mains, 46 pump stations, and sewage tertiary treatment plant located on the island of Malakal. Koror sits over a ridge falling to the coast on either side. For the most part, the commercial sector is located at the crest of the ridge and the residential hamlets stretch down to the coast. Sewage is pumped up to the ridge and flows to the sewage treatment plant (STP). Two other islands, connected by causeways, are also part of the network. The Koror sewerage system network and STP is over was built in the mid 1970's and is in poor condition. The STP was refurbished around 2001 but limited funds for maintenance and limited technical skills mean that much of the plant is unable to operate as designed. The current sewerage flow already exceeds the design capacity of the network and the STP.

26. The poor state of the network and STP leads to system blockages resulting in overflows at the low lying pump stations and discharges of raw sewage to the surrounding environment. The problem is exacerbated by (i) poor maintenance, (ii) an inadequate emergency response capability, (iii) ground water infiltration, and (iv) illegal storm water connections. When overflows occur, the effluent rising through the pump stations discharges through people's yards, basements, taro patches, and into the lagoon. The frequency of overflows is increasing as the network deteriorates and tourist numbers increase.

27. With the exception of Koror, the other 15 state sewerage and wastewater systems are predominately on-site sewerage systems. The Melekeok sewerage treatment works services the Capital Building and the surrounding village. For the majority of states, septic tanks are an appropriate technology for the very low population density. Increases in population in Airai are beginning to create a health and safety problem in that block sizes are declining and the soil types are inappropriate for septic tanks. The major housing development in Airai (Kesebelau) can no longer maintain septic systems without weekly pumping.

28. **Sewerage treatment plants.** The sewage treatment plant in Koror utilizes a settling pond, trickling filters, and finishing ponds before being pumped to an ocean outfall. The plant at Melekeok has reverse osmosis membrane technology to treat the raw waste before it is discharged to the sea. Septic tank standards do exist but there is no mechanism to ensure they are connected properly to the system. Septic tank pumping only takes place if the resident arranges it or the Environmental Quality Protection Board (EQPB) receives a complaint and investigates.

29. **Health issues.** The Belau Hospital records all cases of infectious diseases and has been monitoring the incidence of gastroenteritis since 2008. From 1 April 2011 to 31 March 2012, the hospital recorded 862 cases in Koror⁷, of which 429 (50%) occurred during 3 months from November 2011 to January 2012 when sewerage overflows were the greatest. Hospital records show that during the worst month, 1.25% of Koror's population was treated for gastroenteritis. Given the number of people who do not seek treatment, the actual number of cases is estimated at 6.25% of the total. The risk of a much more serious water borne disease outbreak such as cholera is high. Therefore, the greatest and most immediate concern is the health and safety risks to the people.

⁷ ADB. 2013. *Report and Recommendation to the President, Republic of Palau: Koror-Airai Sanitation Project, Initial Environmental Examination*. Manila.

30. **Impact on economic growth.** The maximum supply of 4 million gallons/day cannot be increased without a large investment in new water supply infrastructure. A new dam is estimated to cost \$30 million. Less expensive alternatives are either not drought-proof or will impact the fresh-water lens that serves as Palau's water source, and have longer-term climate change risks. Alternatively, measures to reduce demand and improve supply system efficiency can lead to the sustainable use of available infrastructure through to 2026. Preparatory assessments indicate that without maintenance investment and pricing controls to dampen demand, the supply will fall to below 72% of demand by 2026.

31. In addition, significant economic costs to Palau are a consequence of the current sewerage network and STP infrastructure. These include (i) direct and indirect health related costs, (ii) overflow clean up costs, and (iii) potential loss of tourism revenue from the impact of a serious disease outbreak. The long-term reputational risk to Palau's tourism industry from reports of a major disease outbreak among its population cannot be underestimated. Fifty percent of Palau's gross domestic product is derived from tourism, mainly from Asian countries which are particularly sensitive to international reports of disease outbreaks.

32. **Financial sustainability.** The government has implemented a policy to achieve full cost recovery for the delivery of water and sewer services by the end of 2015. In 2011 the cost of providing water and sewerage services was approximately \$3.8 million with a government subsidy of 74%. A framework of tariffs and tariff increases to achieve full cost recovery has been agreed and is being implemented. In 2011, the first increase in tariffs was applied since 1986. The tariff structure includes volumetric water and sewer tariffs for domestic and non-domestic connections and includes life line blocks, to protect the poor and vulnerable that are below the cost to treat and deliver water to consumers. Life blocks are on the first 5,000 gallons/month and represent 4.1% of income for those in the 5th quintile of income earners.

33. **Legal and policy framework inadequacy and fragmentation.** Palau's sewerage systems and on-site wastewater treatment and disposal systems are regulated through the Individual Sewage Disposal Act, 1971 (administered by the Ministry of Health), Sewer Use Act, 1984 (administered by the EQPB), and the Environmental Quality Protection Board Regulations Toilet Facilities and Wastewater Disposal Systems Regulations, 1996 (also administered by the EQPB). Through the creation of the Palau Water and Sewer Corporation (PWSC) in September 2011, the Government has integrated water and sewerage services into a single state-owned enterprise responsible for the delivery of water supply and sewerage services. PWSC was subsequently merged into the Palau Public Utilities Corporation (PPUC), Palau's power utility, in June 2013 to form the Palau Public Utilities Corporation Water and Wastewater Operations (WWO).⁸ WWO was declared operationally effective in July 2013. Palau's drinking water quality standards are provided under the EQPB Regulations, 1996 and Public Water Supply Systems Regulations, 1996. These regulations also make provisions for review of the design and development of new water supplies, issue of permits for the operation of water supplies, licensing certification of water supply operators, operation of private water supplies, and powers for enforcing the regulations.

34. Current laws relating to the operation of the public water supply, sewerage treatment and sewage collection systems are generally insufficient to support current operations, and are not a sufficient legislative basis to allow reforms to current water supply arrangements.

35. A recent review of legislative reforms to facilitate water supply infrastructure development and institutional reforms for Palau's public water supplies include (i)

⁸ PWSC was merged into PPUC under the Utilities Consolidation Act 2013 to maximize efficiencies and potential cost savings through shared services and administrative functions.

comprehensive new laws dealing with all aspects of public water supplies; (ii) amendment to the Environmental Quality Protection Act, 1996 to provide a clear mandate for the EQPB to regulate water abstractions following an approval of the National Water Resources Management Plan, and to address issues involving competing claims for rights access water resources; (iii) new regulations made under an amended environmental quality protection act⁹; (iv) a new water resource management policy to manage national and state water resources; (v) legislation to regulate the abstraction of water from the nation's fresh water sources.¹⁰

2. Government's Sector Strategy

36. The goal and actions of the government's medium-term development strategy are (i) investing in water and sanitation, (ii) moving state-owned enterprises toward full cost recovery, and (iii) reducing government expenditure by eliminating subsidies.¹¹ Most of Palau's nation-building projects are nearing completion, and the country's new challenge is taking responsibility for their use and maintenance. ADB assisted Palau in meeting the challenge through the recently completed technical assistance (TA) for a facility for economic and infrastructure management, under which the medium-term development strategy was developed.

37. Capital Improvement Program Office, a division of the Ministry of Public Infrastructure, Industries and Commerce, is responsible for planning and implementing of Palau's public infrastructure projects. The ability of Capital Improvement Program Office to plan and implement new projects is constrained by its small staff. Systems for monitoring, planning, and selecting maintenance and assets need to be improved immediately. Until July 2013, BPW, a division of the Ministry of Public Infrastructure, Industries and Commerce, operated and maintained Palau's water supply and sewage collection and treatment systems. BPW also manages and maintains a wide range of other services, including buildings and roads. Relative to its responsibilities, BPW's budget is too small and erratic, and its productivity low. The government established PWSC and, subsequently, WWO to take over responsibility for the management and delivery of water supply and sewage collection and treatment services.

3. ADB Sector Experience and Assistance Program

38. Palau became a member of Asian Development Bank (ADB) in 2003. ADB assistance has been primarily delivered through technical assistance. ADB supported the development of a medium-term development strategy through TA for the facility for economic and infrastructure management, development of national health insurance scheme, and institutional and policy reforms in water supply and sanitation services. The first lending operations started in 2010, with approval of the Water Sector Improvement Program, that has supported the implementation of institutional and policy reforms to help improve efficiency and sustainability of water supply and sanitation service delivery. This has created an important foundation for the proposed investment project to improve the sanitation systems and services.

39. Official development assistance provides almost all the capital budget, supports improved facilities and services, and supports general economic growth. Japan, Taipei, China, and the United States provide substantial official development assistance to Palau. Japan has granted Palau through its ODA estimated \$15 million to support the upgrading of the water

⁹ Asian Development Bank. 2008. *Technical Assistance to the Republic of Palau for Preparing the Babeldaob Water Supply Project*. Consultant's interim report. Manila.

¹⁰ Government of Palau. 2012. *The Water Policy for the Republic of Palau*. Koror. Government of Palau. 2013. Republic of Palau Public Law 9-44. *The Water Abstraction Act*. Melekeok.

¹¹ Government of Palau. 2009. *Medium-Term Development Strategy: A Plan for Action*. Koror.

supply system. The work is being implemented. Additionally Taipei, China has recently invested in water and sanitation infrastructure.

A. Information and Communications Technology Sector Assessment

1. Sector Performance, Problems, and Opportunities

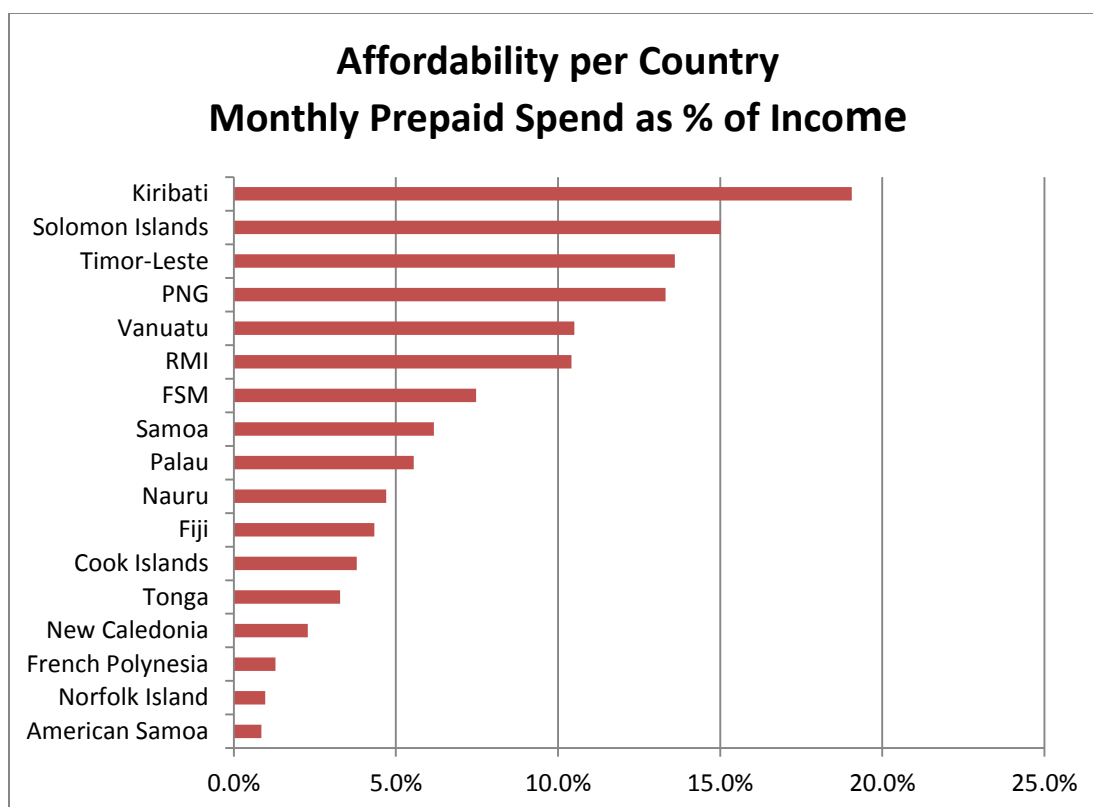
40. Palau relies entirely on geostationary satellite links for international connectivity. Operators purchase capacity on a wholesale basis from major providers such as Intelsat, New Skies and others. At up to \$2,000 per megabit per second (Mbps)/month for a full duplex link, the cost of such satellite capacity is more than three times that achievable with an international submarine cable. A further limitation of such satellites is that their inherent round-trip time delay or latency can cause problems for many interactive internet applications. Satellite services are also susceptible to interruption during cyclones or when wind speeds exceed 100 kilometers per hour. Current international satellite capacities in use by Palau are estimated to total about 172 Mbps.

41. The high cost, limited availability and latency inhibited nature of this international bandwidth represents a major constraint to future development of the ICT sector, and economic growth in Palau more broadly. It is the major factor in widening the ‘digital divide’¹² and a continued drag on improvements in the commercial and day-to-day life of the people. Poor internet quality, even where service is available, means that online commercial transactions and e-services such as distance learning and telemedicine are not achievable. In the case of Palau, whose economy is heavily tourism-dependent, poor ICT services are also a drag on tourism growth.

42. Current mobile service (voice and short messaging service (SMS) only) coverage in Palau is estimated (due to lack of reported data) at 85%, and penetration at 90%. In terms of the affordability of mobile voice services Palau is positioned in the middle range when compared with other Pacific nations, as illustrated below. However to put this into perspective, the United Nations Broadband Commission has targeted that by 2015 the cost of broadband (256 Kbps minimum) should be less than 5% of a country’s per capita gross national income which for Palau would mean reducing the current monthly charge for 256 Kbps from \$650/month to \$45. By comparison, the Cook Islands with similar demographics and which recently moved to an O3b service is able to offer speeds up to 2Mbps for less than \$50/month. Some medium term cost relief and quality improvement has been achieved from the use of O3b Network’s (O3b) medium earth orbit satellite service, which started operating in late 2014, for example Palau National Communication Corporation (PNCC) has doubled it’s bandwidth while keeping the same retail tariff.

¹²

<http://stats.oecd.org/glossary/detail.asp?ID=4719>



Network Strategies Data, June 2013

43. The wholly state-owned PNCC was established in 1982 and provides the standard full range of services including mobile voice and SMS, fixed line telephony, fixed line internet access, and cable TV services. PNCC is the dominant telecommunications operator. PNCC's financial performance is constrained by a \$39 million loan which it took out with the U.S. Rural Utilities Service (USDA/RUS) in 1992. The funds were used to lay fiber-optic cables (the inter-island cables unrepeated offering the potential to expand capacity in the future) throughout Palau and to construct a new switching center. At this time the internet was in its infancy and the project was intended to enable the growth of fixed line voice services. The annual repayments of \$2.3 million represent the major part of PNCC's non-operating expense, and when added to its (unsubsidized) universal service obligations in rural areas, means that PNCC has always struggled to generate a profit. In turn this has impeded its ability to reduce prices and make timely network and technology investments. PNCC has a very limited number of roaming agreements in place, which is an impediment to growing revenues from business visitors and tourists alike. It started providing mobile data services through 3G from Q4 2014. But only very few can afford this costly service, and there is limited coverage while mobile base stations are converted from thin-line microwave to fiber-optic.

44. A second provider, the privately-owned Palau Telecoms, offers broadband wireless internet service, primarily to the hotels, internet cafes, business and Government. Palau Telecoms uses its own wireless links and in some cases capacity leased from PNCC to serve its customers. It estimates that at this time it is able to cover 30% of the population. It operates its own international satellite ground station and leases capacity from wholesale satellite operators. It has not ruled out becoming a mobile operator and has the necessary spectrum licenses in place.

45. A subsidiary of Viamedia Mobile Corporation of Taipei, China, PMC started its operations in 2006 however it suspended operations from July 2014 and remains inoperative, with no clarity as to if, and when, it plans to reopen. Up to this time PMC had provided mobile phone and data services (3G) within a limited coverage area consisting of the main tourist areas and business district of Koror. PMC operated its own international satellite gateway and a number of base stations sufficient to cover its area of service. It also claimed to offer full international roaming service for foreign visitors. The lack of commercial and technical interconnect arrangements with the much larger PNCC was a significant drag on their ability to grow.

46. Currently there are no regulations addressing key ICT market issues such as, interconnection between mobile operators, numbering, tariffs, and competitive behavior.

2. Government's Sector Strategy

47. As a small tourism-driven economy the Government of Palau watched with interest the expansion of submarine cables in the Pacific – including the connection of neighboring Pohnpei State, FSM, and Republic of Marshall Islands with Hawaii. In July 2013 the President of Palau issued Executive Order 344 authorizing the formation of a Palau High Speed Broadband Internet Task Force to pursue the submarine cable project which included the then three operators. In April 2014 an Executive Order passed that mandate to the three Palau appointees to the 'Micro-Pal Fiber Optic Joint Committee' formed by the Federated States of Micronesia (FSM) and Palau governments to oversee the project on behalf of both governments. During its September 2014 mission to Palau ADB upon receipt of stakeholder feedback that the Palau 'task force' be reconstituted as a means of consulting with stakeholders and keeping them informed of the progress of the cable project. The opportunity for Palau to join with the FSM in building the proposed Palau-Yap-Guam cable will secure for the country a future-proof supply of affordable high quality broadband and internet capacity.

48. In November 2013, the Palau Government issued the Palau National ICT Policy setting out its goals, objectives and strategies for ICT for the period 20013 to 2016, one of which is that "All citizens in the Republic of Palau shall have equitable and affordable access to ICT to improve the quality of life through socio-economic development." The policy also calls for the establishment of an independent ICT Regulatory Authority. The Ministry of Public Infrastructure, Industries and Commerce is responsible for implementation of the policy.

49. Government has recognized the need for technical assistance to help build the capacity to implement and manage the needed regulatory reforms and regulatory functions for a modern ICT sector. In particular, the owner and operator-Belau Submarine Cable Corporation (BSCC) of Palau's capacity on the proposed Palau-Yap-Guam cable will effectively have monopoly control over low-cost high-quality international broadband internet capacity for the foreseeable future. It will thus be imperative that adequate regulations exist to ensure that all licensed operators or ISP's in Palau have equal access to broadband capacity in terms of wholesale pricing and interconnect arrangements. In late 2014, the government engaged consultants to develop this regulatory framework. The consultants are drafting an organizational and governance structure for the regulatory office. They are also advising the government to establish necessary legal provision to address issues related to adequate use policy, code of conduct, competition, tariff control etc. in the ICT sector. The office is expected to be in place by Q4-2015.

50. The adoption of these initiatives and the implementation of the regulatory reform are essential for the continued improvement of national ICT infrastructure and the operation of

BSCC on a fair and equitable wholesale basis if the benefits of the proposed submarine cable are to flow through to the people, businesses and government of Palau.

3. ADB Sector Experience and Assistance Program

51. Moving into the new millennium, globalization and digitalization offer challenges and opportunities, particularly for ADB's small and isolated Pacific developing member countries (DMCs). ADB's interim Pacific Approach 2015 prioritizes ICT for the improvement of connectivity among Pacific DMCs and between Pacific DMCs and the rest of the world.¹³ The ADB's Country Operations Business Plan, 2015-2017 for Palau¹⁴, in line with the Strategy 2020 Midterm Review¹⁵, supports the government's goal of developing ICT infrastructure.

52. The regional technical assistance for ICT based inclusive growth and poverty reduction aimed to support regional knowledge sharing and identify new investment opportunities for ICT applications.¹⁶

53. Tonga was successfully connected by a submarine fiber optic cable to the existing global international submarine cable network in 2013 as a result of a jointly-finance ADB and World Bank project. ADB is currently implementing a similar submarine cable project in Solomon Islands and preparing one in Samoa, as well as in the proposed Palau-Yap-Guam cable (these latter two cofinanced by the World Bank). ADB provided technical assistance to support preparatory work in the development of these projects, and will support other Pacific DMCs in assessing the feasibility of future investment options for further international connectivity.¹⁷

54. Promoting competitive pricing in ICT services requires an appropriate regulatory environment. In late 2011 ADB approved technical assistance for the Pacific ICT Regulatory Resource Center, thus contributing in the Pacific to: (i) frequent sharing of experiences and international best practices; (ii) strengthening the capacity of ICT regulatory bodies and policy makers; (iii) providing demand driven advisory services; and (iv) raising broad-based awareness to demand for better ICT regulations.¹⁸ The World Bank is providing follow-on resourcing to the Center.

55. ADB is also supporting projects to identify and implement appropriate ICT applications to support inclusive growth and poverty reduction.

¹³ ADB. 2009. *ADB's Pacific Approach 2010-2014*. Manila.

¹⁴ ADB. 2014. *Palau: Country Operations Business Plan (2015-2017)*. Manila.

¹⁵ ADB. 2014. *Midterm Review of Strategy 2020: Meeting the Challenges of a Transforming Asia and Pacific*. Manila.

¹⁶ ADB. 2009. *Technical Assistance for Information and Communication Technology-Based Inclusive Growth and Poverty Reduction in the Pacific*. Manila.

¹⁷ ADB. 2014. *Regional Technical Assistance for Results-Based Strategy and Sector Planning in the Pacific*. Manila.

¹⁸ ADB. 2014. *Technical Assistance for Strengthening Regulatory Capacity for Information and Communication Technology Development in the Pacific*. Manila.

DEVELOPMENT COORDINATION

A. Major Development Partners: Strategic Foci and Key Activities

56. Palau has a number of development partners—Asian Development Bank (ADB); Australia; European Union (EU); Japan; Taipei,China; United Nations Development Programme (UNDP); the United States and the World Bank (WB). They provide aid through budget support and project or program operations.

57. Australia's strategic focus, through the Partnership for Development 2010–2015, is on capacity development for the education, health and public sectors.

58. The European Union's strategic focus is on the energy and environment sector development especially on renewable energy.

59. The Japan government's strategic focus is on energy, environment, education, and health. Japan supports the improved solid waste management and the establishment of a recycling system, as well as projects for natural resource management and sustainable development. Its support for coral reef research provides an important contribution to monitoring the impacts on sewage outflow on the reef's health. The Government of Japan, through JICA, is supporting a Water Supply Improvement Project, implemented by the Palau Public Utilities Corporation (PPUC).

60. Taipei,China focuses its development assistance on infrastructure, agriculture, and the environment. Taipei,China currently supports small scale rural water collection and transmission infrastructure as well as the cloud services project for the Ministry of Health.

61. The UNDP program focus is on the environment. It mostly supports natural resources management and monitoring. Recently, the UNDP stated its support for a national water resources management policy which incorporates the sanitation sector was provided.

62. The federal grant programs of the government of the United States provides support across all sectors with the major focus on education, health and infrastructure. Through the Compact of Free Association with the United States government, Palau is expected to be able to access infrastructure grants of up to \$40 million over the next 5 years.

63. The World Bank has no projects thus far in Palau; however, it is providing technical assistance towards regulatory reforms of the information and communications technology (ICT) sector. The same TA is supporting the establishment of the proposed Belau Submarine Cable Company, to operate and manage the proposed submarine cable system.

64. The Japan, Taipei,China and United States development assistance programs include volunteer programs. Table 1 summarizes assistance by major development partners.

Table 1 Major Development Partners

Development Partner	Project Name	Duration	Amount (\$ million)
Law, Economic Management, and Public Policy			
ADB	Strengthening Public Sector Management in the North Pacific	2014-2015	0.5
	Medium Term Budget Framework	2009-2012	0.5
	Medium Term Development Strategy	2006-2008	1.0
Information and Communication Technology			
ADB	North Pacific Regional Connectivity Project	2015-2019	25.0
World Bank	Support to ICT regulatory reforms	2014-2015	0.1
Taipei,China	The Health Cloud Project	2014-2015	0.5
Water Supply and Sanitation			
ADB	Water Sector Improvement Program	2010–2015	16.0
	Koror-Airai Sanitation Project		
Japan/JICA	Water Supply Improvement Project	2015-2017	15.0
Taipei, China	Koror Wastewater System Improvements		
	Ngerikiil Water Distribution System		
	Mengellang to Badrulchau Tank Water Transmission	2011–2012	0.3
UNDP/EQPB	Integrated Water Resource Management	2009–2012	0.59
Environment			
Japan	Project for Promotion of Regional Initiative on Solid Waste Management	2011–2015	(...)
	Science and Technology Research Partnership for Sustainable Development		
JICA		2012–2017	3.4
Energy			
European Union	REP-5 (Renewable Energy Program in 5 ACP countries)	2008-2013	2.9
	North-REP	2011-2015	2.3
Japan	Renewable Energy (Solar PV) project	2010	5.0
	Desalination and Renewable Energy project (PEC Fund)	2013	5.0
GEF	Sector Development (UNDP)	2009-2015	1.0
UAE	Renewable Energy	2015	5.0
Danish Government	PIGGAREP Plus	2014-ongoing	0.3
Governments of Italy and Austria	Environment & Sustainable Livelihood through Energy Efficiency & Conservation	2009-ongoing	1.0
US Department of Insular Affairs	Medium and Long Term Energy Sector Planning	2015	0.25
Taipei,China	Solar Systems and LED Street Lighting	2008-2013	4.8

ADB = Asian Development Bank, ACP = African, Caribbean, Pacific, GEF = Global Environment Facility, JICA = Japan International Cooperation Agency, LED = light-emitting diode, PEC = Pacific Environment Community, PIGGAREP = Pacific Islands Greenhouse Gas Abatement through Renewable Energy Project, PV = photovoltaic, REP = renewable energy program, UAE = United Arab Emirates, UNDP = United Nations Development Programme, US = United States

B. Institutional Arrangements and Processes for Development Coordination

65. Development assistance is coordinated through bilateral and multilateral consultations; ADB consults with other development partners when preparing the country operations business plan and exchange information on the projects and technical assistances that are provided to Palau through regular visits to each embassy in Koror, telephone conferences and e-mails. Specifically for the North Pacific Regional Connectivity Project, ADB has participated in a number of joint meetings with the World Bank and the governments of FSM and Palau during project preparation.

66. The Palau Government has created the Office of Budget and Grant Oversight with the goal of providing a roadmap for development partners by setting national priorities. The

Department of Interior has provided a grant to support the Office of Budget and Grant Oversight. The office is in its early stages and is being administered through the leadership of the Adviser in the Office of the Vice-President/Minister of Finance. The goal of the Office of Budget and Grant Oversight is to improve coordination on all grants and technical assistance in Palau. Counterparts for development partners are still fragmented between the Office of the President, Ministry of Foreign Affairs, and Ministry of Finance. The government has not established a formal mechanism for coordination with development partners.

67. With ADB assistance, the government developed a medium-term budget framework. The new budget approach includes all development assistance, cash and in-kind, aligned against each ministry's medium-term development objectives.

C. Achievements and Issues

68. In December 2012, ADB engaged a national development coordination officer to support collaboration between ADB and the government as well as other key stakeholders on the ground including development partners. An ADB development coordination office for Palau was opened in 2014. In 2013 and 2014, ADB worked closely with the government, the private sector and other partners in facilitating the annual national economic summit, which allowed discussion of the key issues and priorities of the government.

D. Summary and Recommendations

69. Development partners are keen to harmonize their development programs for maximum benefit to Palau's socioeconomic development. All major partners agree on the need to prepare a sound, longer-term development strategy that encompasses the RMI's future and transition to the post-Compact era. To optimize development coordination, two areas need particular attention: (i) TA coordination, and (ii) periodic aid consultation meetings chaired by the government.

RISK ASSESSMENT AND RISK MANAGEMENT PLAN

Area of Risk	Risk Level	Management of Risk	Risk Level after Mitigation
The current pace of reform may not be sustained if political support is lost. The risk is that the reform program may stall and progress may be reversed.	High	Senior government officials with support from development partners to develop awareness of the long term objective of the reform programs to garner ongoing support to maintain reform momentum.	Medium
Limited capacity in the public sector to implement programs and ensure sustainability	High	Continue capacity development support by development partners through technical assistance and relevant training programs	Medium
Vulnerable to exogenous shocks due to limited export base and high import dependency	Medium	Economic diversification through private sector development, promoting energy efficiency and renewable energy alternatives.	Low
Climate change risks and exposure to natural disasters	Medium	Continue support to incorporate climate change mitigation into infrastructure projects.	Low
Expiration of compact grants	High	Continue reforms to ensure long term fiscal sustainability	Medium

TA = technical assistance.

Source: Asian Development Bank.